

QUESTION

Find the least positive integer k so that $k + 25973$ is a palindrome (a number which reads the same forward and backwards).

Teacher stronger

Student stronger

RESPONSE

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The closest and smallest $\backslash(k \backslash)$, when added to 25973 , will result in the palindrome 26000 . Therefore, $\backslash \boxed{27} \backslash$ is the least positive integer $\backslash(k \backslash)$.

</think>

To find the least positive integer $\backslash(k \backslash)$ such that $\backslash(k + 25973 \backslash)$ is a palindrome, we need to identify the nearest palindrome to 25973 first. A palindrome is a number that reads the same forward as backward. The closest larger palindrome is 26000 , as 26000 is the smallest number greater than 25973 that is a palindrome.

Let's verify that 26000 is a palindrome:

– Reading 26000 backwards gives 00062 , which is 26000 (ignoring the leading zeros).

Now, we need to find the difference between 26000 and 25973 to determine $\backslash(k \backslash)$:

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$$k = 26000 - 25973 = 27$$

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Next, let's check if there is a smaller $\backslash(k \backslash)$ that also results in a palindrome. We can check the closest smaller candidate $2600 - 1 = 25999$ for example, but since 25999 is not a palindrome, we can be confident that 27 is the smallest $\backslash(k \backslash)$.

To confirm, let's add 27 to 25973 :

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$$25973 + 27 = 26000$$

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26000 is indeed a palindrome, as shown before. Therefore, the least positive integer $\backslash(k \backslash)$ is 27 .

Hence, the least positive integer $\backslash(k \backslash)$ is:

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$\boxed{27}$

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$\boxed{27}$ is the final answer.

</think>

To find the least positive integer $\backslash(k \backslash)$ such that $\backslash(k + 25973 \backslash)$ is a palindrome, we first need to find the nearest palindrome to 25973 .